MTH316

Examination - January Semester 2024

Multivariable Calculus

Friday, 19 April 2024 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FIVE (5)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1a

Define a function $f : \{(x, y) \in \mathbb{R}^2 \mid x \neq \sqrt{y^2 + 9}\} \longrightarrow \mathbb{R}$ by

$$f(x, y) = \frac{xy}{x - \sqrt{y^2 + 9}}.$$

Question 1a(i)

Determine

$$\lim_{(x,y)\to(-1,4)} f(x, y),$$

if it exists. Otherwise, show that it does not exist.

(3 marks)

Question 1a(ii)

Show that

$$\lim_{(x,y)\to(3,0)} f(x, y)$$

does not exist.

(8 marks)

Question 1b

Define a function $f : \{(x, y) \in \mathbb{R}^2 \mid x \neq -y\} \longrightarrow \mathbb{R}$ by

$$f(x, y) = \begin{cases} \dfrac{k\left(x^2 - y^2\right)}{(x + y)^2}, & (x, y) \neq (0, 0), \\ 1, & (x, y) = (0, 0). \end{cases}.$$

Determine the value of k, if any, for the function to be continuous at (0, 0).

(9 marks)

Question 2

A solid region E is described by

$$E : \left\{(x, y, z) \mid -1 \leq y \leq x + 1,\ x^2 + z^2 \leq 1\right\}.$$

Question 2a

Sketch the projections of region E onto the three coordinate planes: $x = 0$, $y = 0$ and $z = 0$.

(6 marks)

Question 2b

Set up an integral in Cartesian coordinates that gives the volume of region E. Do not evaluate the integral.

(4 marks)

Question 2c

Convert the integral in Question 2b to one in cylindrical coordinates and evaluate it for the volume of solid E.

(10 marks)

Question 3

A jet of air impinges on a rectangular plate and causes changes in the air pressure acting on the surface of the plate. The air pressure (in kPa) away can be modelled by the function

$$P(x, y) = 2x^3 + y^2 - 6xy + 30,$$

where x and y are distances measured (in cm) from one corner of the plate surface and satisfies $0 \leq x \leq 5, 0 \leq y \leq 10$.

Question 3a

Calculate the gradient of the pressure function at (3, 2). Hence or otherwise, find a vector **V** in the direction along which the pressure is locally constant at the point (3, 2).

(6 marks)

Question 3b

An ant crawling on the plate moves from (3, 2) to (0, 5) in a straight line. Determine the gradient of air pressure experienced by the ant in its direction of movement at (3, 2).

(4 marks)

Question 3c

Apply derivative tests to determine the lowest and highest air pressures acting on the plate surface and their corresponding locations. State any theorem used.

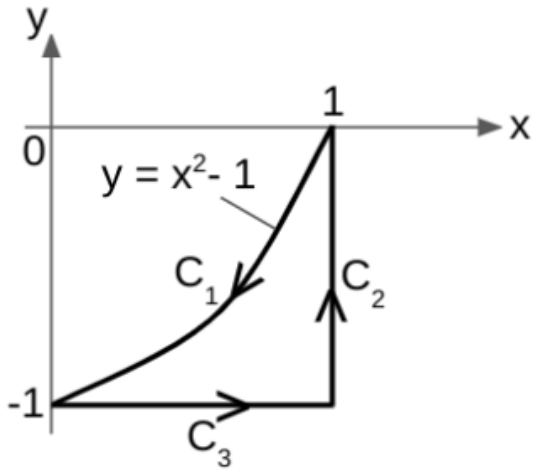
(10 marks)

Question 4

Define a force field $F(x, y)$ by

$$\mathbf{F}(x, y) = \begin{bmatrix} y \\ 4x^2 \end{bmatrix}.$$

Three paths C_1, C_2 and C_3 that are connected to form a closed path are shown below.



<u>Figure Q4</u>

Question 4a

Compute directly the work done by $F(x, y)$ along the path C_1.

(7 marks)

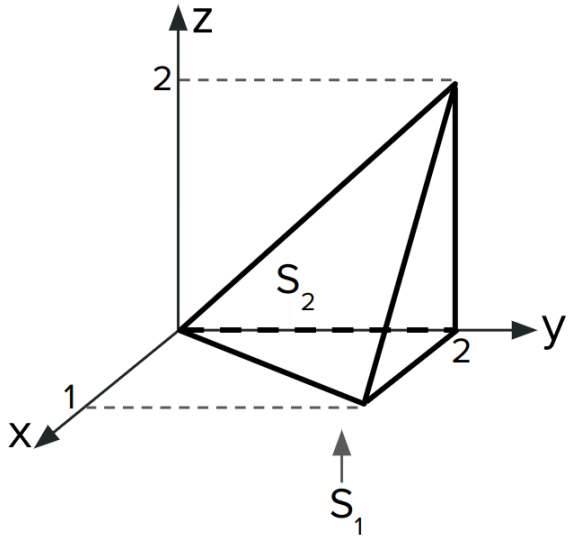
Question 4b

Use Green's theorem to evaluate the work done by $F(x, y)$ along the combined path $C_2 + C_3$.

(13 marks)

Question 5

A vector field $\mathbf{F}(x, y, z)$ and a tetrahedron with a triangular base are defined below. All four surfaces of the tetrahedron are orientated with outward normal vectors.



$$\mathbf{F}(x, y, z) = \begin{bmatrix} y \\ xz - 3y \\ z \end{bmatrix}.$$

Question 5a

Surface S_1 is the surface of the tetrahedron on the xy-plane bounded by $x = 0$, $y = 2$ and $y = 2x$. Evaluate the flux of $\mathbf{F}(x, y, z)$ through S_1.

(5 marks)

Question 5b

Surface S_2 is the plane $z = y - 2x$ on the tetrahedron bounded by $x = 0$, $y = 2$ and $y = 2x$. Evaluate the flux of $F(x, y, z)$ through S_2.

(10 marks)

Question 5c

Using the divergence theorem, evaluate the flux of $\mathbf{F}(x, y, z)$ through the other two surfaces of the tetrahedron (outside of S_1 and S_2).

(5 marks)

----- END OF PAPER -----